

YOUCHENG ZONG

 Shenyang, China

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 25 years old

 Industrial LLMs, industrial time-series modeling, intelligent metallurgy, and decision support



Education

2024.09-Present	Northeastern University	Control Science and Engineering	Ph.D. Student
2021.09-2024.06	Anhui University of Technology	Materials and Chemical Engineering	M.Eng.
2017.09-2021.06	Anhui University of Technology	Automation	B.Eng.

Academic Achievements

Journal Publications

- **First author.** LLM-Driven Human-AI Collaborative Decision Support System for Complex Industrial Processes: A Case Study in Metallurgy. *Neural Networks*, 2026. [DOI](#).
- **First author.** A meta-contrastive learning hybrid model for adaptive temperature trend prediction in variable ladle preheating. *Engineering Applications of Artificial Intelligence*, 2025. [DOI](#).
- **First author.** Hybrid Grid Search and Bayesian optimization-based random forest regression for predicting material compression pressure in manufacturing processes. *Engineering Applications of Artificial Intelligence*, 2025. [DOI](#).
- **First author.** Iron-Tapping State Recognition of Blast Furnace Based on Bi-GRU Composite Model and Post-Processing Classifier. *IEEE Sensors Journal*, 2023. [DOI](#).
- **First author.** Judgment of blast furnace iron-tapping status based on data differential processing and dynamic window analysis algorithm. *Progress in Natural Science: Materials International*, 2023. [DOI](#).
- **Second author.** Zero-Shot Capillary Segmentation in Dermoscopy Images via SAM2: A Case Study on Oral Mucosa. *IEEE Journal of Biomedical and Health Informatics*, 2025. [DOI](#).
- **Second author.** Influence of heavy reduction during solidification process of billets based on 3D reconstruction of dendrites. *Journal of Iron and Steel Research International*, 2025. [DOI](#).

Patents and Software Copyrights

- Invention patentFirst inventor Published A method for judging the iron-tapping state of blast furnace hot-metal ladles.
- Invention patentFirst inventor Published A pressure prediction method and system for high-performance material compression processes.
- Invention patentThird inventor Published A device and method for synchronous detection of reduction amount and pressure value in continuous-casting soft reduction.
- Utility model First inventor Granted An intelligent temperature and humidity regulation system for poultry incubation based on a mobile app.
- Software copyrightFirst owner Granted Ten software copyrights, including systems for continuous-casting billet weighing monitoring, intelligent scrap batching, and team collaboration platforms.

Project Experience

- › Industry-university-research project on intelligent transformation of a Baowu Changgang plant; responsible for equipment selection, communication-network construction, algorithm design, and software development.
- › Industry-university-research project on process-control efficiency improvement for a Baowu Magang LF furnace process; responsible for equipment selection, communication-network construction, algorithm design, and software development.
- › General Program of the National Natural Science Foundation of China; participated in data-driven mechanism analysis for a rheological-process-related research topic.
- › Young Scientists Fund of the National Natural Science Foundation of China; participated in data-driven mechanism analysis for large-ingot solidification research.
- › National Key Research and Development Program project; participated in endpoint-prediction software development for a 5G-enabled beneficiation and smelting process.